

Skill Arena - Phase 4 Document 5

Security Operations Center (SOC), Incident Response, Threat Intelligence & Vulnerability Management Architecture

Document Purpose

Define the operational security framework used to monitor, detect, investigate and respond to security threats.

Business Objective

Protect platform availability, user accounts, treasury assets and operational integrity through continuous security operations.

Security Operations Center

Central command center responsible for security visibility and threat response.

SOC Responsibilities

Monitoring, Detection, Investigation, Containment, Recovery and Post-Incident Analysis.

Security Monitoring Framework

Collect and analyze logs, alerts, authentication events, infrastructure events and application activity.

Threat Intelligence Platform

Aggregate internal and external threat indicators for proactive defense.

Threat Intelligence Sources

Security vendors, infrastructure telemetry, application monitoring and incident history.

Alert Management System

Categorize and prioritize security events based on severity and impact.

Severity Levels

Informational, Low, Medium, High, Critical and Emergency.

Incident Response Framework

Preparation, Detection, Analysis, Containment, Eradication, Recovery and Lessons Learned.

Incident Classification

Account Compromise, Fraud Event, Infrastructure Attack, Data Exposure, Service Disruption and Insider Threat.

Investigation Workflow

Alert → Triage → Investigation → Escalation → Resolution → Documentation.

Forensic Analysis

Preserve logs, evidence, replay records and system state for investigations.

Threat Hunting Program

Proactive search for hidden threats, suspicious activity and emerging attack patterns.

Vulnerability Management

Continuous identification, assessment and remediation of security weaknesses.

Vulnerability Scanning

Automated scans across applications, infrastructure and services.

Patch Management

Track vulnerabilities, prioritize fixes and validate remediation.

Penetration Testing

Regular testing of applications, APIs, infrastructure and security controls.

Security Dashboards

Threat Overview, Active Incidents, Vulnerability Status, Risk Trends and Security Health.

Executive Security Reporting

Provide leadership with security posture, risks, incidents and remediation progress.

Business Continuity Integration

Coordinate with disaster recovery and operational resilience plans.

Audit Requirements

All investigations, alerts and security actions permanently recorded.

Compliance Integration

Support audits, regulatory reviews and security certifications.

Access Controls

SOC tools restricted to authorized security personnel.

Performance Requirements

Security monitoring and response capabilities operate continuously 24/7.

Acceptance Criteria

Platform can rapidly detect, investigate, contain and recover from security incidents.